

## Supplementary Material: Failure-Aware Smartphone Inertial Dead Reckoning Under GNSS Outages

**Article Title:** Failure-Aware Smartphone Inertial Dead Reckoning Under GNSS Outages: An Empirical Study of Learned Velocity, Sensor Fusion, Map Constraints, and Distribution Shift

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### Overview

This supplementary document provides extended tabular data, detailed per-trip forensic metrics, hardware profiling logs, and mathematical derivations supporting the main article. All evaluations utilize the public IO-VNBD benchmark dataset (Onyekpe et al., 2021) and on-device Android mobile execution on a Google Pixel 7a smartphone.

### Supplementary Table S1: Per-Trip Performance Breakdown Under 60-Second GNSS Blackouts

The main manuscript presents aggregate and category-level metrics across the 13 usable held-out test routes. Table S1 provides the comprehensive route-by-route results for both unassisted strapdown inertial dead reckoning and the proposed NavSense adaptive fusion engine.

Table S1. Detailed route-by-route evaluation across all 13 usable held-out test trips under 60-second satellite outages.

| Trip ID | Route Environment | Reference Distance (m) | Pure Strapdown Drift (m) | Pure Strapdown Drift (%) | NavSense Adaptive Drift (m) | NavSense Adaptive Drift (%) | SIH Benchmark Status (<10%) | Observed Dynamic Characteristics                                   |
|---------|-------------------|------------------------|--------------------------|--------------------------|-----------------------------|-----------------------------|-----------------------------|--------------------------------------------------------------------|
| Vta21   | Suburban Town     | 786.7                  | 406.8                    | 51.45%                   | 61.5                        | 7.88%                       | MET (PASS)                  | Steady suburban cruising along arterial corridor; bounded drift    |
| Vta22   | Suburban Town     | 641.3                  | 176.7                    | 27.72%                   | 107.7                       | 16.84%                      | Unmet                       | Frequent stop-and-go queue; velocity dampens with traffic          |
| Vta23   | Suburban Town     | 570.1                  | 352.9                    | 62.40%                   | 156.8                       | 26.16%                      | Unmet                       | Sharp 90-degree intersection turn during blackout window           |
| Vta24   | Suburban Town     | 384.8                  | 293.6                    | 73.42%                   | 57.0                        | 14.65%                      | Unmet                       | Commercial zone corridor; low-speed transient manoeuvres           |
| Vta26   | Suburban Town     | 258.7                  | 292.8                    | 574.65%                  | 66.3                        | 47.31%                      | Unmet                       | Short transit crawl; large percentage due to short travel distance |

| Trip ID             | Route Environment    | Reference Distance (m) | Pure Strapdown Drift (m) | Pure Strapdown Drift (%) | NavSense Adaptive Drift (m) | NavSense Adaptive Drift (%) | SIH Benchmark Status (<10%) | Observed Dynamic Characteristics                                 |
|---------------------|----------------------|------------------------|--------------------------|--------------------------|-----------------------------|-----------------------------|-----------------------------|------------------------------------------------------------------|
| <b>Vta27</b>        | Suburban Town        | 818.9                  | 306.5                    | 37.18%                   | 95.9                        | 12.68%                      | Unmet                       | Signalised arterial road with moderate speed transitions         |
| <b>Vta28</b>        | Suburban Town        | 565.3                  | 281.8                    | 64.92%                   | 120.6                       | 25.30%                      | Unmet                       | Roundabout entry and deceleration into residential zone          |
| <b>Vw12</b>         | Mountain Route       | 1500.2                 | 650.3                    | 43.40%                   | 45.3                        | 3.01%                       | <b>MET (PASS)</b>           | Continuous mountain descent; highly dynamic centripetal cues     |
| <b>Vw14a</b>        | Mountain Route       | 1511.1                 | 312.6                    | 21.06%                   | 130.4                       | 8.54%                       | <b>MET (PASS)</b>           | Rural mountain corridor with sweeping curves and steady throttle |
| <b>Vw14b</b>        | Mountain Route       | 1264.5                 | 432.8                    | 42.48%                   | 121.0                       | 10.19%                      | Unmet                       | Consecutive hairpins; centripetal acceleration transitions       |
| <b>Vw16a</b>        | Mountain Route       | 899.9                  | 388.5                    | 51.76%                   | 136.1                       | 16.88%                      | Unmet                       | High curvature hill climb with frequent throttle modulations     |
| <b>V-Vfa02</b>      | Motorway             | 1460.3                 | 313.9                    | 25.44%                   | 157.2                       | 11.69%                      | Unmet                       | Unaccelerated high-speed cruising; negative TCN regression bias  |
| <b>Vw15</b>         | Stationary Control   | 1.3                    | 5.6                      | 450.48%                  | 0.82                        | 63.83%                      | Unmet                       | Zero-velocity control trip; standstill clamped by causal ZVD     |
| <b>Mean / Total</b> | <b>All 13 Routes</b> | <b>820.2</b>           | <b>324.21</b>            | <b>89.66%</b>            | <b>96.65</b>                | <b>16.76%</b>               | <b>3 / 13 (23.08%)</b>      | <b>Complete Held-Out 60-s Evaluation Set</b>                     |

*Note: In accordance with the experimental protocol, trips with total durations shorter than 70.0 seconds are excluded from the 60-second evaluation set because they cannot accommodate the 10-second pre-outage calibration window plus the 60-second blackout duration. Macro-average normalized drift weights each dynamic trip equally, excluding the stationary control trip Vw15 from the percentage mean per Equation (13) to avoid distortion from the near-zero travel-distance denominator. Normalised drift percentages are calculated using the reference distance travelled during the corresponding 60-s blackout window,  $S_i(T)$ , as defined in Eq. (13). The*

*'Reference Distance' column reports the route/reference distance associated with the evaluation record and is not necessarily identical to the blackout-window travel distance.*

#### Supplementary Table S2: Spectral Vibration Speed Audit Across 64 Vehicle Trips

To evaluate the hypothesis that vehicle road and engine vibration captured by consumer smartphone IMUs provides a reliable proxy for vehicle forward speed, Table S2 summarizes the spectral census across all 64 passenger-car trips in the IO-VNBD dataset (>450,000 one-second FFT windows).

Table S2. Cross-trip spectral vibration census across four standardized road categories.

| Road Environment            | Trips Analyzed (N) | Total 1-s Windows  | Dominant Spectral Peak (Hz) | Pearson Correlation ( $r$ ) with Speed | Spearman Correlation ( $\rho$ ) with Speed | Physical Interpretation                                                                 |
|-----------------------------|--------------------|--------------------|-----------------------------|----------------------------------------|--------------------------------------------|-----------------------------------------------------------------------------------------|
| <b>Suburban Towns (Vta)</b> | 30                 | 218450             | $2.24 \pm 0.18$ Hz          | -0.028                                 | -0.024                                     | Consistent with low-frequency vehicle-body/chassis dynamics                             |
| <b>Dense Urban (Vtb)</b>    | 12                 | 40890              | $2.31 \pm 0.22$ Hz          | -0.041                                 | -0.035                                     | Consistent with road-surface excitation and low-frequency vehicle dynamics              |
| <b>Mountain Routes (Vw)</b> | 20                 | 170920             | $2.45 \pm 0.15$ Hz          | -0.019                                 | -0.016                                     | Consistent with chassis/body dynamics under curved-road motion                          |
| <b>Motorways (V-Vfa)</b>    | 2                  | 39180              | $2.86 \pm 0.12$ Hz          | -0.038                                 | -0.031                                     | Persistent spectral peak across speed bands; physical mechanism not directly identified |
| <b>Complete Corpus</b>      | <b>64</b>          | <b>&gt;450,000</b> | <b>2.2–2.5 Hz</b>           | <b>-0.032</b>                          | <b>-0.028</b>                              | <b>No reliable speed signal identified at 10 Hz feature rate</b>                        |

*Methodological Note: Power spectral density was computed causally using Welch periodograms across 1.0-second sliding windows (10 samples, zero-padded to 64 points) on tri-axial specific forces and angular velocities.*

#### Supplementary Table S3: On-Device Android Execution Profiling and Numerical Parity

Table S3 details the execution latency and cross-platform numerical parity between the mobile Android deployment (Java/EJML/ONNX Runtime) on a Google Pixel 7a and the 64-bit Python reference engine.

Table S3. Mobile execution benchmarks and cross-platform numerical parity on Google Pixel 7a.

| Pipeline Subsystem                    | Implementation Language | Runtime Framework                    | Mean Execution Latency | Peak Latency (99th %tile) | Maximum Discrepancy vs. Python 64-bit Engine |
|---------------------------------------|-------------------------|--------------------------------------|------------------------|---------------------------|----------------------------------------------|
| <b>IMU Ingestion &amp; Decimation</b> | Kotlin                  | Android SensorManager HAL            | 0.12 ms                | 0.35 ms                   | Identical (Zero difference)                  |
| <b>Coordinate Leveling &amp; PCA</b>  | Java                    | Native Linear Algebra                | 0.24 ms                | 0.58 ms                   | Leveling Angle: <0.005°                      |
| <b>TCN Forward Inference</b>          | C++ / Java Bindings     | ONNX Runtime Mobile v1.16            | 5.82 ms                | 7.84 ms                   | Speed: <0.001 m/s                            |
| <b>15-State ESKF Propagation</b>      | Java                    | Efficient Java Matrix Library (EJML) | 0.94 ms                | 1.38 ms                   | Position: 0.24177 m; Heading: 0.01829°       |
| <b>Decoupled Velocity Damping</b>     | Java                    | Custom EJML Matrix Slice             | 0.18 ms                | 0.32 ms                   | Velocity: 0.02117 m/s                        |

| Pipeline Subsystem                 | Implementation Language | Runtime Framework                  | Mean Execution Latency | Peak Latency (99th %tile) | Maximum Discrepancy vs. Python 64-bit Engine |
|------------------------------------|-------------------------|------------------------------------|------------------------|---------------------------|----------------------------------------------|
| <b>Open-Loop Vector Mapping</b>    | Kotlin                  | OSMDroid Offline Engine            | 1.85 ms                | 3.42 ms                   | Render: <1 frame                             |
| <b>Total Navigation Epoch Loop</b> | <b>Mixed</b>            | <b>End-to-End Android Pipeline</b> | <b>9.15 ms</b>         | <b>13.89 ms</b>           | <b>Within the 100-ms epoch budget</b>        |

### Supplementary Section S1: Mathematical Derivation of Causal Receptive Field Reach

The causal Temporal Convolutional Network (TCN) operates over a sliding temporal buffer of  $W = 100$  samples (10.0 s at 10 Hz). To ensure that predictions depend strictly on past motion dynamics with zero dependency on future epochs, causal convolutions are enforced via asymmetric left-padding.

For a 1D convolutional network with  $L$  dilated residual blocks, kernel size  $k$ , and dilation factor  $d_l$  at layer  $l$ , the mathematical receptive field reach  $R$  (expressed in sample steps) is given by:

$$R = 1 + \sum_{l=0}^{L-1} (k - 1) \cdot d_l$$

In the TCN architecture, the parameters are:

- \* Number of dilated residual layers:  $L = 4$
- \* Convolutional kernel size:  $k = 3$
- \* Dilation schedule:  $d_l \in \{1, 2, 4, 8\}$  for  $l \in \{0, 1, 2, 3\}$

Each residual block contains two sequential dilated convolutional layers with identical dilation  $d_l$ . Substituting these values:

$$R_{\text{block},l} = 2 \cdot (k - 1) \cdot d_l = 2 \cdot (3 - 1) \cdot d_l = 4d_l$$

Summing across all four dilated blocks:

$$R = 1 + \sum_{l=0}^3 4d_l = 1 + 4 \cdot (1 + 2 + 4 + 8) = 1 + 4 \cdot 15 = 61 \text{ samples}$$

At the nominal 10.0 Hz sampling rate, 61 samples correspond to an effective causal look-back interval of:

$$\tau_{\text{causal}} = (61 - 1) \times 0.10 \text{ s} = 6.0 \text{ seconds}$$

representing input dependency on the causal interval  $[t - 60, t]$ .